Lab 6 : Dynamic Routing Configurations : RIP, EIGRP, OSPF, BGP

Objectives :

- To grasp how routers automatically discover remote networks and maintain routing tables without manual entry of every path.
- To configure and analyse the operation of RIP, EIGRP, OSPF, and BGP, focusing on how different algorithms (Distance Vector, Link-State, and Path Vector) automatically discover network topologies and populate routing tables.

Theory :

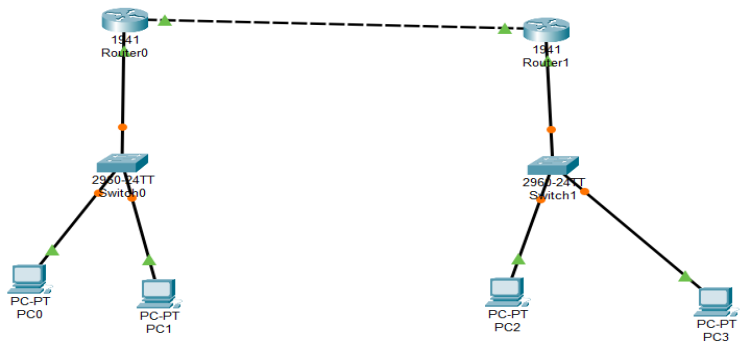
Dynamic routing protocols allow routers to automatically discover remote networks and build their own routing tables. Unlike static routing, these protocols can adapt to topology changes without manual intervention.

1. **Routing Information Protocol (RIP) :**

   RIP is one of the oldest distance vector protocols. It is characterized by its simplicity but has limitations in larger networks.

   - Algorithm: Uses the Distance Vector algorithm.
   - Metric: Uses Hop Count to determine the best path, with a maximum limit of 15 hops.
   - Operation: Routers send their entire routing table to their neighbours every 30 seconds.
   - Versions: RIP v1 is classful (no subnetting support), while RIP v2 is classless and supports CIDR.

Network Topology :



Configuration :

For PCs :

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers :

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0	10.0.0.1
				gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0	10.0.0.2
				gig 0/1	192.168.2.1

Commands :

Router 0 :

router rip

version 2

network 192.168.1.0

network 10.0.0.0

auto-summary

Router 1 :

router rip

version 2

network 192.168.2.0

network 10.0.0.0 no

no auto-summary

Observation :

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.3

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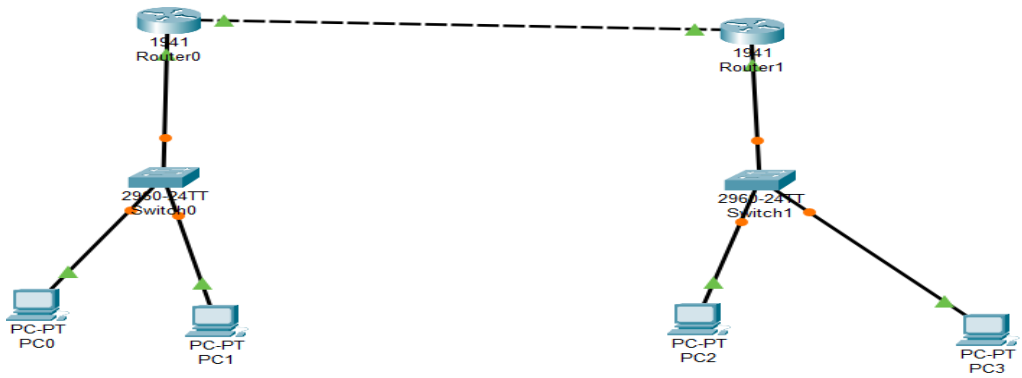
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. **Enhanced Interior Gateway Routing Protocol (EIGRP) :**

   EIGRP is a hybrid protocol that combines the best features of both distance vector and link-state protocols.

   - Algorithm: Uses the Diffusing Update Algorithm (DUAL) to ensure loop-free paths.
   - Metric: Uses a composite metric based on Bandwidth and Delay (optionally Reliability and Load).
   - Operation: Routers only send updates when a topology change occurs, which saves bandwidth.
   - Feature: Supports unequal cost load balancing, which is unique among these protocols. **Network Topology :**



Configuration :

For PCs :

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PC0	192.168.1.2	255.255.255.0	192.168.1.1
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For Routers :

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0	10.0.0.1
				gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0	10.0.0.2
				gig 0/1	192.168.2.1

Commands :

Router 0 :

router eigrp 100

network 192.168.1.0 0.0.0.255

10.0.0.0 0.0.0.255

no auto-summary

Router 1 :

router eigrp 100

network 192.168.2.0 0.0.0.255 network

network 10.0.0.0 0.0.0.255

no auto-summary

Observation :

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
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Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

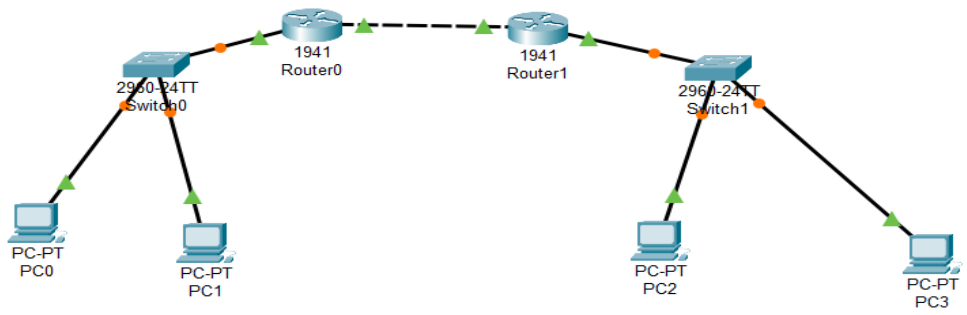
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3. **Open Shortest Path First (OSPF) :**

   OSPF is a widely used link-state routing protocol designed for scalability and fast convergence.

   - Algorithm: Uses Dijkstra's Shortest Path First (SPF) algorithm.
   - Metric: Uses Cost, which is calculated based on reference bandwidth divided by interface bandwidth.
   - Operation: Routers discover neighbors using "Hello" packets and exchange Link State Advertisements (LSAs) to build a synchronized topology database.
   - Advantage: Offers very fast convergence compared to RIP.

Network Topology :



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Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
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Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0	10.0.0.2
				gig 0/1	192.168.2.1

Commands :

Router 0 :

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.255 area 0
```

Router 1 :

```
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.255 area 0
```

Observation :

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

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Ping statistics for 192.168.1.2:
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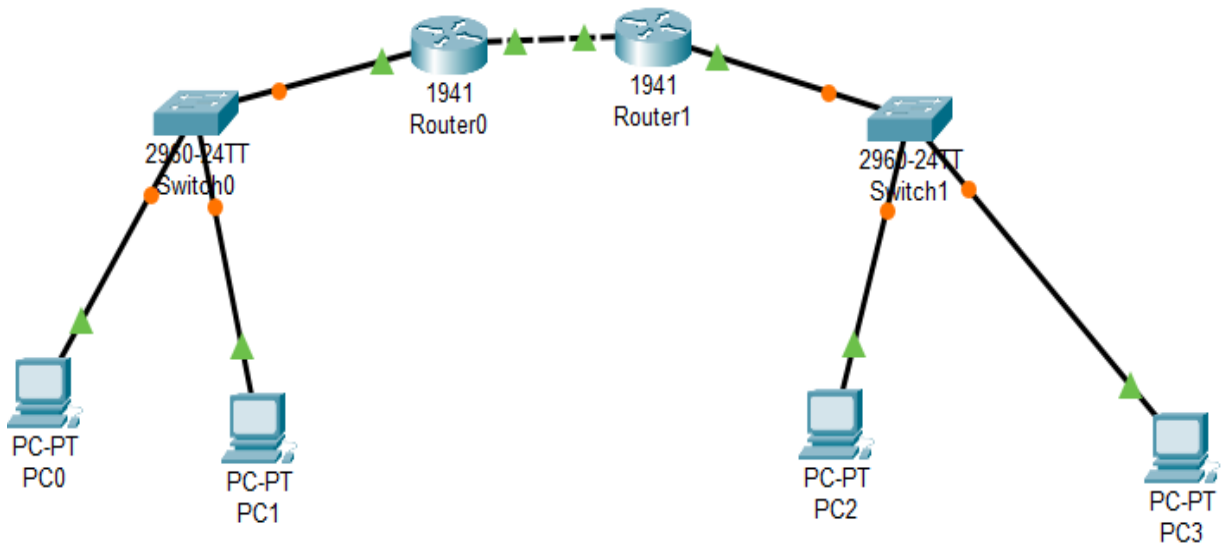
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. **Border Gateway Protocol (BGP) :**

   BGP is a path vector protocol and is considered the "routing protocol of the internet".

   - Purpose: It is used to connect different Autonomous Systems (AS), which are groups of networks under a single administration.
   - Operation: BGP uses path attributes (like AS-Path) rather than simple metrics to select the best path.
   - Reliability: It uses TCP port 179 for reliable communication between routers.
   - Scalability: It is highly reliable and scales to manage the entire global internet routing table.

Network Topology :



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For Routers :

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0	10.0.0.1
				gig 0/1	192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0	10.0.0.2
				gig 0/1	192.168.2.1

Commands :

Assumption :

R1 -> AS 65001

R2 -> AS 65002

Router 0 :

```
router bgp 65001
neighbor 10.0.0.2 remote-as 65002
network 192.168.1.0 mask 255.255.255.0
```

Router 1 :

```
router bgp 65001
neighbor 10.0.0.1 remote-as 65001
network 192.168.2.0 mask 255.255.255.0
```

Observation :

```
C:\>ping 192.168.1.3

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Result :

During this lab, we replaced static routing with dynamic routing by configuring four different protocols: RIP, EIGRP, OSPF, and BGP. After the setup, the routers automatically exchanged routing information with one another, eliminating the need to manually define every route. We confirmed successful configuration by examining the routing tables, where newly learned routes were displayed. Finally, end-to-end connectivity was validated by successfully pinging devices across the entire network.

Discussion :

In this lab, we configured multiple dynamic routing protocols and tested each to understand how they differ. RIP was the easiest to configure, but it is limited because it selects routes based only on hop count. OSPF appeared more advanced, as it determines the best path by considering link cost, which is influenced by bandwidth. EIGRP responded the fastest when network changes were introduced, showing quick convergence. BGP demonstrated how large-scale networks operate by exchanging routing information between network groups rather than focusing on individual routers. Finally, ping tests were performed to verify overall network connectivity.

Conclusion :

In this lab various algorithms of dynamic routing were successfully implemented.